Fiscal Year:	FY 2008	**Task Last Updated:**	FY 04/14/2009
PI Name:	Perchonok, Michele Ph.D.		
Project Title:	Comparative Packaging Study		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Johnson Space Center		
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PI Web Page:	http://hefd.jsc.nasa.gov/		
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2007	**End Date:**	09/30/2008
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Woolford, Barbara	**Contact Phone:**	218-483-3701
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE PI changed to Patricia Catauro (briefly was Thomas Oziomek but then Catauro took over in the same fiscal year) in FY2009. See Catauro for FY2009 and beyond.		
COI Name (Institution):	Oziomek, Thomas (MEI Technologies Inc., Houston, TX)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	Future long duration manned space flights beyond low earth orbit will require the food system to remain safe, acceptable and nutritious. Development of high barrier food packaging will enable this requirement by preventing the ingress and egress of gases and moisture. New high barrier food packaging materials have been identified through a trade study. Practical application of this packaging material within a shelf life test will allow for better determination of whether this material will allow the food system to meet given requirements after the package has undergone processing. The reason to conduct shelf life testing, using a variety of packaging materials, stems from the need to preserve food used for mission durations of several years. Chemical reactions that take place during longer durations may decrease food quality to a point where crew physical or psychological well-being is compromised. This can result in a reduction or loss of mission success. The rate of chemical reactions, including oxidative rancidity and staling, can be controlled by		

Task Description:	limiting the reactants, reducing the amount of energy available to drive the reaction, and minimizing the amount of water available. Water not only acts as a media for microbial growth, but also as a reactant and means by which two reactants may come into contact with each other. Two food products, dry cereal and peanuts, and one ingredient, cottonseed oil, susceptible to oxygen and water vapor ingress were packaged in three different packaging materials and incubated at a fixed temperature (72?F) and varying relative humidity. Storage at three relative humidities (25%, 50%, 75%) will simulate the midpoint and potential ranges of humidities on the Orion vehicle. Relative humidity is an important variable since it may impact the barrier properties of certain materials. At three month intervals, sensory and quantitative analysis are being performed on the food items to measure changes due to oxygen and moisture migration through the packaging. Free fatty acid (FFA) and peroxide value (PV) are being performed on the oil for quantification of oxidative rancidity. Moisture analysis quantifies the moisture ingress of the cereal and hexanal analysis quantifies the level of rancidity in the peanut samples. Sensory methods, used only for cereal and peanuts, include Difference from Control testing which indicates when the food has changed enough for consumers to notice a difference. The data associated with the three packaging materials will be compared in order to determine which is best suited for future exploration missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2008. NOTE PI changed to Patricia Catauro (briefly was Thomas Oziomek but then Catauro took over in the same fiscal year) in FY2009. See Catauro for FY2009 and beyond.
Bibliography Type:	Description: (Last Updated: 01/30/2012)